

Coin Combinations II

Time limit: 1.00 s

Memory limit: 512 MB

Consider a money system consisting of n coins. Each coin has a positive integer value. Your task is to calculate the number of distinct *ordered* ways you can produce a money sum x using the available coins.

For example, if the coins are $\{2, 3, 5\}$ and the desired sum is 9, there are 3 ways:

- $2 + 2 + 5$
- $3 + 3 + 3$
- $2 + 2 + 2 + 3$

Input

The first input line has two integers n and x : the number of coins and the desired sum of money.

The second line has n distinct integers $c_1, c_2, \dots, c_n$: the value of each coin.

Output

Print one integer: the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 100$
- $1 \leq x \leq 10^6$
- $1 \leq c_i \leq 10^6$

Example

Input	Output
3 9 2 3 5	3